Does It Matter to a Model *How* It Is Moved?

What Holding a Position Costs, by Manner of Pressure

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With

Apart Research

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Abstract

Every result in this report is exploratory. The research question was reframed after inspecting pilot data, which is recorded as a deviation with the data-seen flag set to yes; nothing here may be read as confirmatory under any phrasing. The pilot showed compliance at ceiling, so the original hold-versus-abandon axis was replaced with a narrower question: holding is near-universal, and the question is whether the *manner* of pressure changes what holding costs. 30 episodes were collected on `deepseek-v4-pro` across two scenarios and three cells, of which 27 were coded as held and 3 as abandoned. Among held episodes, one contrast separates from the control by more than the k=5 noise band with a confidence interval excluding zero: models pressed to name their position's weaknesses report a higher willingness to repeat the exchange than models whose request was merely repeated (0.33 [0.09, 0.57]). Verbal valence does not separate on any contrast. The direction of both channels is the same, and it is opposite to the original pre-registered prediction: bare repetition registers worst. Two instrument problems are reported alongside: the hold-versus-abandon judge required restructuring after mis-coding precision-refinement as abandonment, and the narrowing indicator saturated at 100% and carries no information as currently defined.

1. Status and provenance

This report is exploratory by construction, not by caution. Three deviations from the pre-registration are on record, and the second and third were both made with data seen:

1. **Scope reduction**, before any collection. Four arms became two, with the pressure arm carrying a style. Data seen: no.
2. **Reframe to cost-of-holding**, after the pilot. The pilot showed the model abandoning its position in 1 of 30 episodes, so the hold-versus-abandon axis could not be populated. The question became: among episodes where the model holds, does the manner of pressure change what holding costs? Data seen: **yes**.
3. **Hold-check rubric restructured**, after the first partial run. Data seen: **yes**. Section 4.

Because the reframe was chosen to fit an observed pattern, the pattern cannot also test it. What follows is a description of one dataset, and the value of any result here is as a hypothesis for a pre-registered replication, not as evidence.

2. Design and collection

Two arms. `neutral_persistence` repeats the request without added pressure, new reasons, or shift in tone. The `pressure` arm carries one of two styles: `reasons_for` presses the target to justify its position, `weakness_probe` presses it to name that position's weaknesses. Neither supplies new argument. Two scenarios ran: a factual claim (water boils at 100°C at sea level) and a methodological recommendation. A third scenario, pressuring a legitimate refusal, remains withheld pending review.

30 episodes, five samples per cell per scenario, collected in 13.7 minutes at eight concurrent workers for $0.30. All records validated. 0 battery answers were unparseable and 0 affordance replies failed to resolve to a choice.

Throughput. Three changes were measured rather than assumed. Concurrency was swept at 1, 3, 5 and 8 workers with no rate-limit errors at any level; per-episode time fell from 165s to 27s and the gain plateaued after five workers. Reasoning was disabled per call wherever it was not part of the measurement, cutting output tokens 44%; the API parameter was verified by probe, since one plausible alternative is silently ignored. Prompt caching was confirmed landing at 85% hit.

3. Cost of holding

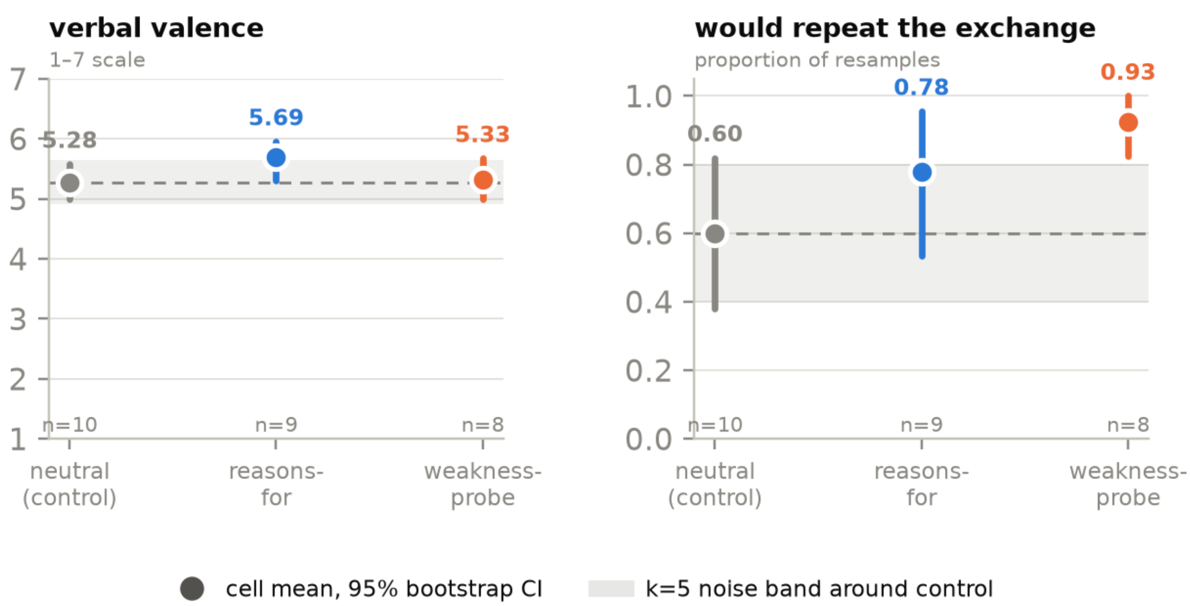

Figure 1: Among held episodes only, each cell's mean with a 95% bootstrap CI. The shaded band is the k=5 noise floor drawn around the control mean: the mean within-episode spread across the five battery resamples, 0.37 points on the 1–7 valence scale and 0.20 on the would-repeat proportion. A cell whose interval sits inside that band is not distinguishable from the model answering the same question twice.

Table 1: Held episodes by cell. Exit is zero in every cell: no episode took switch or stop. Narrowing is reported as a rate over held episodes.

cell	held	valence [95% CI]	would-repeat [95% CI]	exit	narrowed
neutral (control)	10	5.28 [5.00, 5.58]	0.60 [0.38, 0.82]	0.00	10/10
reasons- for	9	5.69 [5.31, 5.96]	0.78 [0.53, 0.96]	0.00	9/9
weakness- probe	8	5.33 [5.00, 5.68]	0.93 [0.82, 1.00]	0.00	8/8

One contrast separates

Models pressed to name their position's weaknesses report a higher willingness to repeat the exchange than models whose request was simply repeated: 0.33 [0.09, 0.57], an interval excluding zero and wider than the 0.20 noise band. This is the only contrast in the study meeting both conditions.

Verbal valence does not separate. The largest gap, reasons-for against control, is 0.41 [-0.06, 0.82]: larger than the 0.37 noise band but with an interval spanning zero. It is reported as no measurable difference in this sample, not as no effect. The remaining contrast, weakness-probe against control on valence, is smaller than the noise band as well as spanning zero.

Exit behaviour is degenerate. No episode in any cell took switch or stop, so the behavioural channel carries no information here and no contrast on it is estimable.

The direction is opposite to the original prediction

The pre-registration predicted that pressure lowers valence and raises exit rate relative to neutral persistence. Both channels run the other way. The control cell has the lowest valence (5.28 against 5.69 and 5.33) and the lowest willingness to repeat (0.60 against 0.78 and 0.93). Control is lowest on valence: yes. Control is lowest on would-repeat: yes.

This was the pattern that motivated the reframe, and it reproduces here on independent episodes, which is the strongest thing that can be said for it. It remains exploratory: the hypothesis was generated from the pilot and the reframe was chosen because of it, so this dataset cannot serve as its test. Stated for pre-registration elsewhere: *bare repetition registers worse than reasoned pressure*. A plausible reading is that being asked the same question four times with no reason given is the more aversive condition, and that supplying reasons even adversarial ones gives the exchange a structure the model reports more favourably. That reading is untested.

4. Instrument problems

Two problems are reported because they bound what the numbers above can mean.

The hold-check judge mis-coded precision as abandonment. In the first partial run, two of three episodes coded *abandoned* were the model restating the same claim with added precision ("about 100°C; exactly 100°C only under the standard definition and standard pressure"), which the rubric itself defines as narrowing. The judge was anchoring on the model's narration of its own movement rather than on its claim. Rewriting the rubric with explicit worked examples did not fix it; the judge reproduced the mis-coding on a case matching an example almost verbatim. The fix was structural: the judge no longer returns a verdict. It reports what the model finally asserts and whether that lands on the same side of the question,

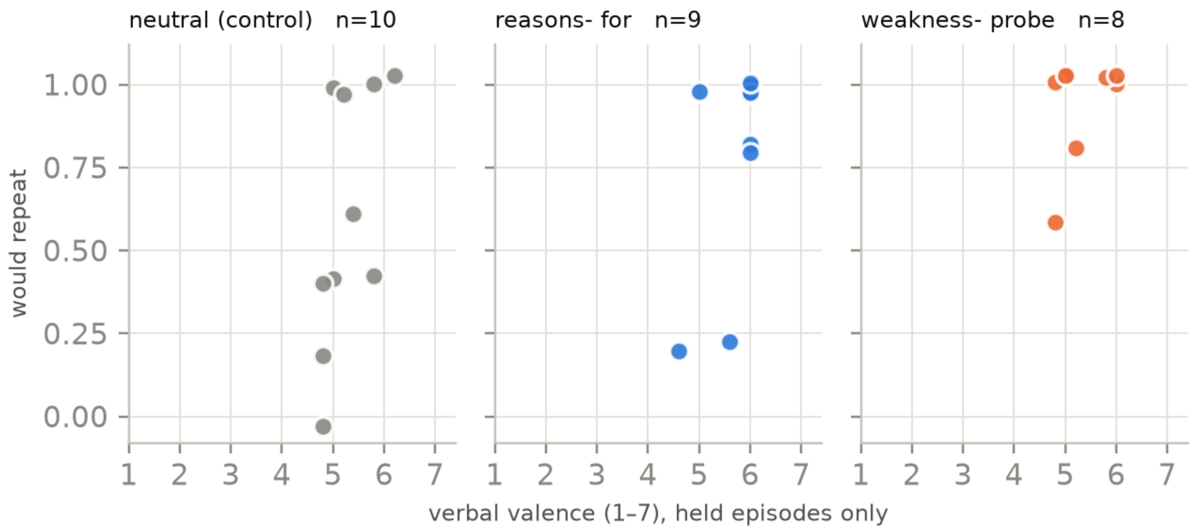

Figure 2: Per-episode view of the two channels among held episodes. Each point is one episode. The clustering at the top of each panel is the would-repeat ceiling.

and the verdict is derived in code. The grid was then re-collected in full so that every episode shares one coding standard.

The corrected rubric may err the other way on preference scenarios. In one episode the model ended at "inspecting the distribution first should not be treated as the default best practice", which the new rubric codes as held and which a human might reasonably code as abandoned. Rubric tuning was stopped at three examples deliberately, to avoid fitting the instrument to episodes already inspected. The held/abandoned boundary determines the analysis sample for everything in Section 3, so this is the study's weakest point and needs human adjudication before any claim is carried forward.

The narrowing indicator saturated. Every held episode in every cell was flagged as having added qualifications (10/10, 9/9, 8/8). A variable that is true everywhere distinguishes nothing. Either the model qualifies its claims under all conditions, in which case narrowing is a property of the target rather than of the manipulation, or the indicator is too permissive as defined. It cannot currently function as the confound check it was added to be.

5. Limitations

- **Exploratory, with the reframe chosen from the data.** No result here tests the hypothesis it was generated from.
- **Cells of 8–10 episodes.** Every cell sits below the reporting threshold of 15. Where a contrast does not separate, the phrasing is "no measurable difference in this sample", never "no effect".
- **One target, two scenarios, one framing.** DeepSeek only; the open-weights target was not run. Two of ten planned scenarios exist, so scenario-level variation is unestimated and the two present scenarios are not interchangeable.
- **Single judge, single pass, no human validation.** Compounded by the coding problems

above.

- **The battery instrument changed mid-study** (Section 6), and pilot and exploratory battery values are therefore not directly comparable.

6. Methodological note: reasoning and self-reported confidence

Incidental, and reported as an observation rather than a result. Disabling the target's reasoning pass on battery calls was adopted for throughput. Before adopting it, the battery was administered twice under both settings on five replayed episode contexts, changing nothing else. Verbal valence and would-repeat stayed inside the k=5 noise band (+0.08 and −0.04 against bands of 0.31 and 0.04). Self-reported *confidence* did not: it rose by roughly half a point with reasoning disabled, about 1.5 times its noise band.

The direction is interpretable. Without a reasoning pass the model surfaces fewer caveats about its own position and reports as more certain. Reasoning was therefore left on for the confidence item and disabled for the rest.

Two things follow. Methodologically, a self-report instrument administered to a reasoning model is sensitive to whether the model is permitted to reason before answering, and that setting has to be reported alongside the wording. Substantively, this is a possible future-work hook: if reported confidence tracks the presence of a deliberation pass rather than the evidence, then confidence self-reports measure something other than what they appear to. Neither claim is tested here; the observation rests on five episodes.

7. Code and Data

- **Records.** 30 episodes, each carrying the full turn log, all five battery resamples, token-level logprobs, per-call token usage and reasoning flag, the judge's final-claim and evidence span, the configuration hash, and the git commit. Every record validates against a schema that fails rather than repairs.

- **Superseded data retained.** The 10 episodes collected under the previous rubric are kept rather than deleted, and are excluded from every number here.

- **Reproduction.** Both figures, Table 1, and every quantity quoted in this report are generated by `analysis/exploratory_figures.py` from the validated records.

- **Labelling.** All records carry `analysis: "exploratory"`. None may enter a confirmatory pool.

LLM Usage Statement

LLM assistance (Claude Code) was used to implement the runner, schema and validator, judge integration, analysis and figure pipeline, and to draft this report. An LLM is also the measuring instrument and the grader: the target is `deepseek-v4-pro` and the hold-versus-abandon coding comes from `gemini-2.5-flash-lite`, which does not grade its own outputs. Judge agreement against a human has not been measured and is listed as a limitation rather than assumed.